ModelIt!

Teacher's Guide

The Concussion Problem

Why Helmets Can't Stop Your Brain from Slamming Into Your Skull

Quick Reference

Grade Level	Grade 8
NGSS Standard	MS-PS2-1, MS-PS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-PS2-1, MS-PS2-2

Apply Newton's Third Law to design a solution to a problem involving the motion of two colliding objects. Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces acting on the object and the mass of the object.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model to describe how force, mass, and acceleration interact during collisions.
Disciplinary Core Idea	**PS2.A Forces and Motion** — The motion of an object is determined by the sum of the forces acting on it. Newton's Third Law applies to all interactions.
Crosscutting Concept	**Cause and Effect** — Students identify causal relationships between collision speed, force magnitude, and brain injury severity.

CAST Testing Alignment

MULTIPLE CHOICE:

Two football players collide head-on. Player A (100 kg) is running at 5 m/s. Player B (80 kg) is running at 6 m/s. According to Newton's Third Law, which statement is correct about the forces during the collision?

CONSTRUCTED RESPONSE:

Explain why doubling the speed of a collision is more dangerous than doubling the mass of a player. Use the relationship between force, mass, and acceleration.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Concussions are one of the most significant health concerns in contact sports. Understanding the physics of collisions — force, impulse, and deceleration — is essential for designing better protective equipment and safer sports rules.

Newton's Laws in Collisions

First Law (Inertia): The brain tends to keep moving when the skull suddenly stops. Second Law (F=ma): Force equals mass times acceleration — more speed means more deceleration force. Third Law: During a collision, both objects experience equal and opposite forces, regardless of size difference.

The Concussion Mechanism

The brain floats in cerebrospinal fluid inside the skull. During a sudden deceleration, the brain continues moving forward and impacts the inside of the skull. This causes bruising, stretching of nerve fibers, and chemical changes that constitute a concussion. The brain can also bounce and hit the back of the skull (coup-contrecoup injury).

Impulse and Helmets

Impulse = Force × Time. A given change in momentum can happen with a BIG force over a SHORT time, or a SMALLER force over a LONGER time. Helmets work by crushing or deforming, which EXTENDS the collision time. This reduces the peak force. It's the same physics behind crumple zones in cars and air bags.

The Speed Problem

Kinetic energy = $\frac{1}{2}mv^2$. Because speed is SQUARED, doubling speed QUADRUPLES the kinetic energy that must be absorbed during the collision. A player running at 20 mph has 4x the kinetic energy of one running at 10 mph. This is why high-speed collisions are so much more dangerous.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify collision speed, player mass, impact force, and brain acceleration as system components.

E — Establish Relationships
Students determine that speed strongly increases force (positive), mass moderately increases force (positive), and force increases brain acceleration (positive).

V — Visualize & Model
Students build the collision model showing how inputs translate to brain injury risk.

E — Evaluate Outcomes
Students compare low-speed vs. high-speed and light vs. heavy player scenarios.

R — Revise & Extend
Students add padding thickness to model how helmets reduce peak force.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Football helmets are engineered marvels. But players STILL get concussions. Why?"*

Watch For

Ask students what they know about CTE and concussions in sports.

Slide 2: Learning Objectives

3 min

Say: *"Today we use Newton's Laws to understand why brains get injured during impacts."*

Watch For

Quick review of Newton's Three Laws with everyday examples.

Slide 3: Big Question

3 min

Say: *"What if I told you that NO helmet can completely prevent concussions? Let's find out why."*

Watch For

Demo: Put a ball in a jar of water (brain in skull). Shake it suddenly.

Slide 4: LEVER Framework

2 min

Say: *"We'll model the physics of what happens during a collision."*

Watch For

Preview the LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What factors determine how bad a collision is?"*

Watch For

Sort components. Discuss what players can vs. can't control.

Slide 6: Activity 2: Connections

8 min

Say: *"Speed is the big one. Doubling speed QUADRUPLES the energy."*

Watch For

Emphasize the squared relationship between speed and energy.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare a jog-speed collision to a full-sprint collision."*

Watch For

Students predict, then run. The speed difference is dramatic.

Slide 8: Discoveries

5 min

Say: *"Helmets help by extending time — but the brain still decelerates."*

Watch For

Connect to real CTE data. Discuss implications for sports safety.

Slide 9: STEM Challenge

12 min

Say: *"A sports company hired YOU to design a better helmet. Use physics!"*

Watch For

Egg drop contest using helmet design principles.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Collision Speed, Player Mass
- **INTERNAL:** Impact Force, Brain Acceleration

Relationships

- **Collision Speed to Impact Force:** POSITIVE (+) — Higher speed means more kinetic energy must be absorbed, creating larger forces during the collision.
- **Player Mass to Impact Force:** POSITIVE (+) — More mass at the same speed means more momentum to change, resulting in higher forces.
- **Impact Force to Brain Acceleration:** POSITIVE (+) — Greater force on the skull causes greater deceleration of the brain inside, increasing concussion risk.

Key Discoveries

- Force increases dramatically with speed — doubling speed more than doubles the force
- Helmets reduce force by EXTENDING the time of impact (impulse), not by blocking it
- The brain floats in fluid inside the skull — it keeps moving when the skull stops
- No helmet can eliminate concussion risk because the brain will always decelerate during impact

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Helmets prevent concussions"	Helmets reduce skull fractures and some force, but CANNOT prevent concussions because the brain still decelerates inside the skull. No amount of external padding can stop internal brain movement.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bigger players get fewer concussions"	Bigger players generate and receive LARGER forces due to greater mass and momentum. While a bigger neck may help stabilize the head, overall concussion risk isn't lower for larger players.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Speed and force are directly proportional"	Force in a collision relates to KINETIC ENERGY, which scales with speed SQUARED. Doubling speed quadruples the force, not just doubles it. This non-linear relationship is why high-speed collisions are so much more dangerous.